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GATE ECE · COMMUNICATIONS · CHAPTER-WISE PYQS

Random Signals and Noise

GATE ECE Communications · Previous Year Questions with Answer Key

56	36	1987 – 2025	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	16 questions
Marks 2	39 questions
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Answer key	at the end

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Marks 1 16 questions

Q1. GATE ECE 2024

A white Gaussian noise $w(t)$ with zero mean and power spectral density $\frac{N_0}{2}$, when applied to a first-order RC low pass filter produces an output $n(t)$. At a particular time $t = t_k$, the variance of the random variable $n(t_k)$ is _____.

Open solution

(A) $\frac{N_0}{4RC}$ (B) $\frac{N_0}{2RC}$ (C) $\frac{N_0}{RC}$ (D) $\frac{2N_0}{RC}$

Q2. GATE ECE 2023

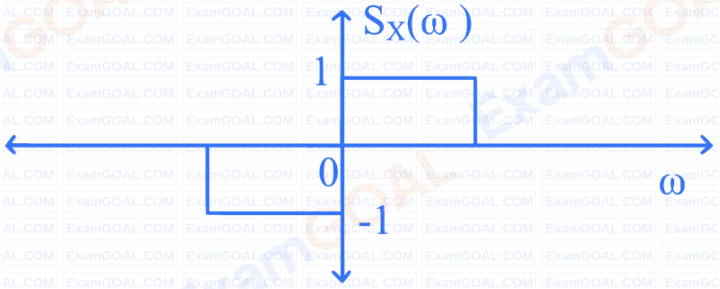
For a real signal, which of the following is/are valid power spectral density/densities?

Open solution

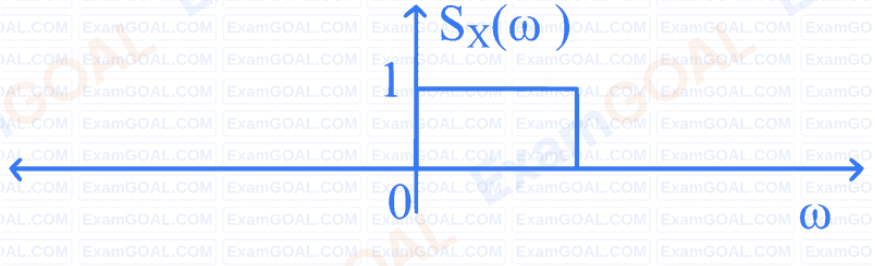
(A) $S_x(\omega) = \frac{2}{9+\omega^2}$

(B) $S_x(\omega) = e^{-\omega^2}\cos^2\omega$

(C)



(D)



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Q3. GATE ECE 2022

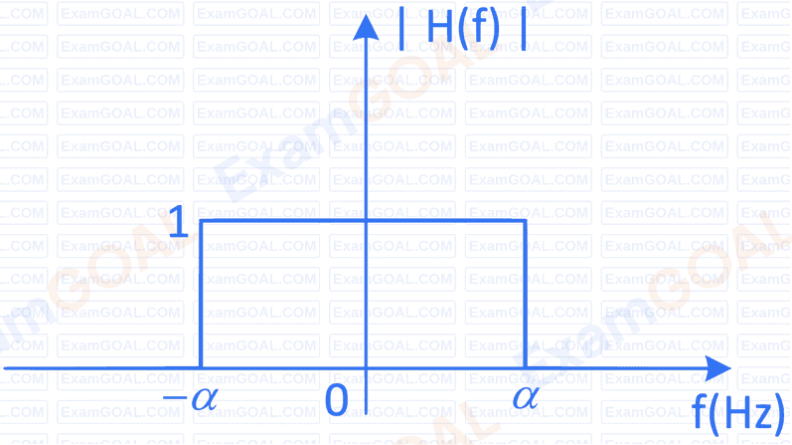
Open solution

The frequency response H(f) of a linear time-invariant system has magnitude as shown in the figure.

Statement I : The system is necessarily a pure delay system for inputs which are bandlimited to $-\alpha \leq f \leq \alpha$.

Statement II : For any wide-sense stationary input process with power spectral density $S_X(f)$, the output power spectral density $S_Y(f)$ obeys $S_Y(f) = S_X(f)$ for $-\alpha \leq f \leq \alpha$.

Which one of the following combinations is true?



(A) Statement I is correct, Statement II is correct

(B) Statement I is correct, Statement II is incorrect

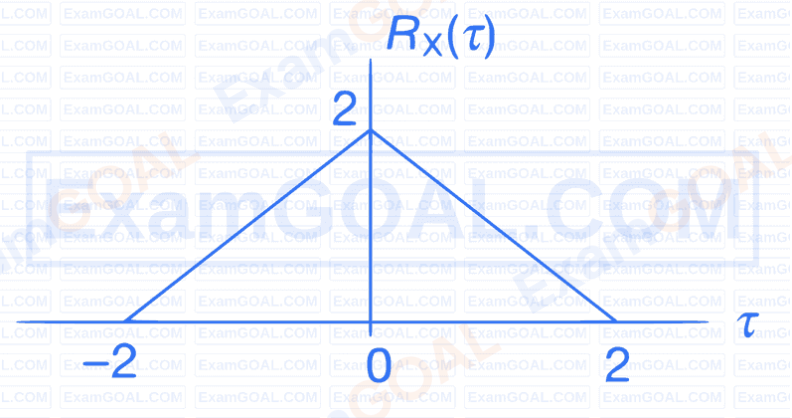
(C) Statement I is incorrect, Statement II is correct

(D) Statement I is incorrect, Statement II is incorrect

Q4. GATE ECE 2021

Open solution

The autocorrelation function $R_X(\tau)$ of a wide-sense stationary random process $X(t)$ is shown in the figure.



The average power of $X(t)$ is _____

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Q5. GATE ECE 2020

The random variable

$Y = \int_{-\infty}^{\infty} W(t)\phi(t)dt, \quad \text{where } \phi(t) = \begin{cases} 1, & 5 \le t \le 7 \\ 0, & \text{otherwise} \end{cases}$

and $W(t)$ is a real white Gaussian noise process with two-sided power spectral density $S_W(f) = 3\ \mathrm{W/Hz}$, for all f. The variance of Y is ____ .

Open solution

Q6. GATE ECE 2017 Set 2

Consider the random process
x(t) = U + Vt.
Where U is a zero mean Gaussian random variable and V is a random variable uniformly distributed between 0 and 2. Assume that U and V are statistically independent. The mean value of the random process at t = 2 is _____

Open solution

Q7. GATE ECE 2014 Set 4

If calls arrive at a telephone exchange such that the time of arrival of any call is independent of the time of arrival of earlier or future calls, the probability distribution function of the total number of calls in a fixed time interval will be

(A) Poisson (B) Gaussian (C) Exponential (D) Gamma

Open solution

Q8. GATE ECE 2012

Two independent random variable X and Y are uniformly distributed in the interval [- 1, 1]. The probability that max [X, Y] is less than 1/2 is

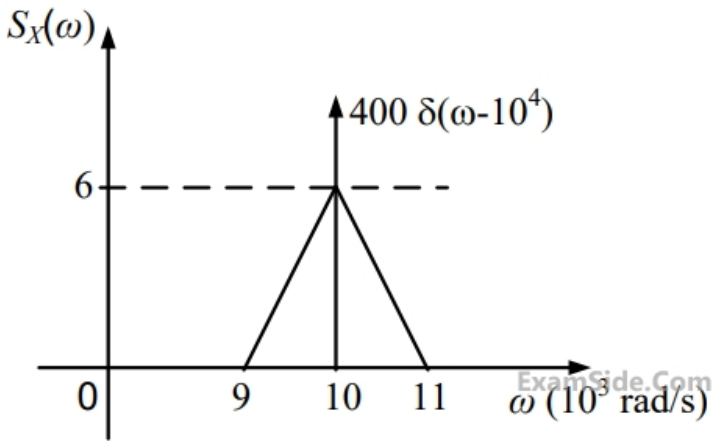
(A) 3/4 (B) 9/16 (C) 1/4 (D) 2/3

Open solution

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Q9. GATE ECE 2012

The power spectral density of a real process X(t) for positive frequencies is shown below. The value of $E\left[X^2(t)\right]$ and $E\left[X(t)\right]$, respectively, are



(A) $6000/\pi,\ 0$

(B) $6400/\pi,\ 0$

(C) $6400/\pi,\ 20\left(\pi\sqrt{2}\right)$

(D) $6000/\pi,\ 20\left(\pi\sqrt{2}\right)$

Open solution

Q10. GATE ECE 2007

If E denotes expectation, the variance of a random variable X is given by

(A) $E\left[X^2\right]-E^2[X]$

(B) $E\left[X^2\right]+E^2[X]$

(C) $E\left[X^2\right]$

(D) $E^2[X]$

Open solution

Q11. GATE ECE 2001

The PDF of a Gaussian random variable X is given by $p_x(x)=\frac{1}{3\sqrt{2\pi}}\exp\left[-(x-4)^2/18\right]$.

The probability of the event {X = 4} is

(A) 1/2

(B) $1/\left(3\sqrt{2\pi}\right)$

(C) 0

(D) 1/4

Open solution

Q12. GATE ECE 1998

The amplitude spectrum of a Gaussian pulse is

(A) uniform

(B) a sine function

(C) Gaussian

(D) an impulse function

Open solution

Q13. GATE ECE 1998

The ACF of a rectangular pulse of duration T is

(A) a rectangular pulse of duration T

(B) a rectangular pulse of duration 2T

(C) a triangular pulse of duration T

(D) a triangular pulse of duration 2T

Open solution

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Q14. GATE ECE 1998

The probability density function of the envelope of narrow band Gaussian noise is

(A) poisson (B) Gaussian (C) Rayleigh (D) Rician

Open solution

Q15. GATE ECE 1997

A probability density function is given by $p(x) = K \exp(-x^2/2), -\infty < x < \infty$.
The value of K should be

(A) $1/\sqrt{2\pi}$ (B) $\sqrt{2/\pi}$ (C) $1/2\sqrt{\pi}$ (D) $1/\pi\sqrt{2}$

Open solution

Q16. GATE ECE 1995

For a narrow band noise with Gaussian quadrature and inphase components, the probability density function of its enveolope will be

(A) uniform (B) Gaussian (C) exponential (D) Rayleigh

Open solution

Marks 2 39 questions

Q17. GATE ECE 2025

Consider a real-valued random process

$$f(t) = \sum_{n=1}^{N} a_n p(t - nT),$$

where $T > 0$ and N is a positive integer. Here, $p(t) = 1$ for $t \in [0, 0.5T]$ and 0 otherwise. The coefficients a_n are pairwise independent, zero-mean unit-variance random variables. Read the following statements about the random process and choose the correct option.

(i) The mean of the process $f(t)$ is independent of time t.

(ii) The autocorrelation function $E[f(t)f(t+\tau)]$ is independent of time t for all τ. (Here, $E[\cdot]$ is the expectation operation.)

(A) (i) is TRUE and (ii) is FALSE (B) Both (i) and (ii) are TRUE

(C) Both (i) and (ii) are FALSE (D) (i) is FALSE and (ii) is TRUE

Open solution

Q18. GATE ECE 2024

A source transmits a symbol s, taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise channel. The received noisy symbol r is given by $r = s + w$, where the noise w is zero mean with variance 4 and is independent of s.

Using $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{t^2}{2}} dt$, the optimum symbol error probability is _____.

(A) $\frac{2}{3}Q(2)$ (B) $\frac{4}{3}Q(1)$ (C) $\frac{2}{3}Q(1)$ (D) $\frac{4}{3}Q(2)$

Open solution

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Q19. GATE ECE 2024

Open solution

Let $X(t) = A\cos(2\pi f_0 t + \theta)$ be a random process, where amplitude A and phase θ are independent of each other, and are uniformly distributed in the intervals $[-2, 2]$ and $[0, 2\pi]$, respectively. $X(t)$ is fed to an 8-bit uniform mid-rise type quantizer.

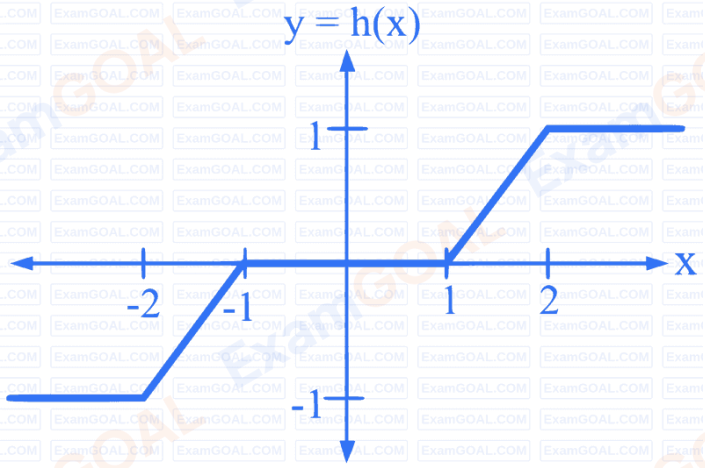
Given that the autocorrelation of $X(t)$ is $R_X(\tau) = \frac{2}{3}\cos(2\pi f_0 \tau)$, the signal to quantization noise ratio (in dB, **rounded off to two decimal places**) at the output of the quantizer is _____.

Q20. GATE ECE 2023

Open solution

A random variable X, distributed normally as N(0, 1), undergoes the transformation Y = h(X), given in the figure. The form of the probability density function of Y is

(In the options given below, a, b, c are non-zero constants and g(y) is piece-wise continuous function)



(A) $a\delta(y-1) + b\delta(y+1) + g(y)$

(B) $a\delta(y+1) + b\delta(y) + c\delta(y-1) + g(y)$

(C) $a\delta(y+2) + b\delta(y) + c\delta(y-2) + g(y)$

(D) $a\delta(y+2) + b\delta(y-2) + g(y)$

Q21. GATE ECE 2023

Open solution

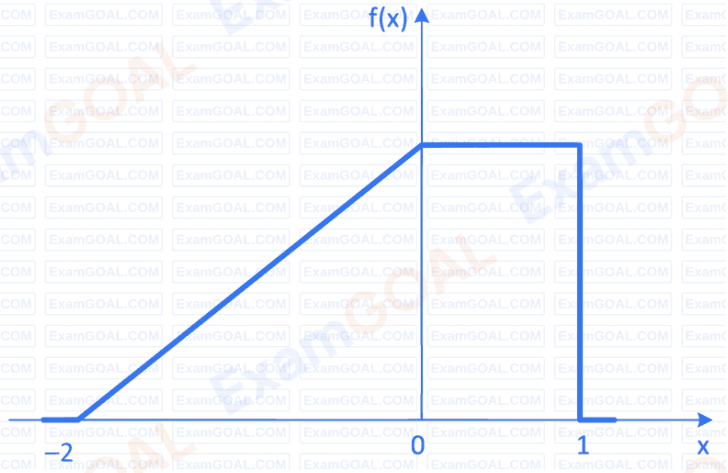
Let X(t) be a white Gaussian noise with power spectral density $\frac{1}{2}$W/Hz. If X(t) is input to an LTI system with impulse response $e^{-t}u(t)$. The average power of the system output is _____ W (rounded off to two decimal places).

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Q22. GATE ECE 2022

Open solution

Consider a real valued source whose samples are independent and identically distributed random variables with the probability density function, f(x), as shown in the figure.



Consider a 1 bit quantizer that maps positive samples to value α and others to value β. If α^* and β^* are the respective choices for α and β that minimize the mean square quantization error, then $(\alpha^* - \beta^*)$ = _____ (rounded off to two decimal places).

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Q23. GATE ECE 2017 Set 1

Open solution

Let $X(t)$ be a wide sense stationary random process with the power spectral density $S_x(f)$ as shown in figure (a), where f is in Hertz (Hz). The random process $X(t)$ is input to an ideal low pass filter with the frequency response

$$H(f) = \begin{cases} 1, & |f| \leq \frac{1}{2} Hz \\ 0, & |f| > \frac{1}{2} Hz \end{cases}$$

As shown in Figure (b). The output of the low pass filter is $y(t)$.

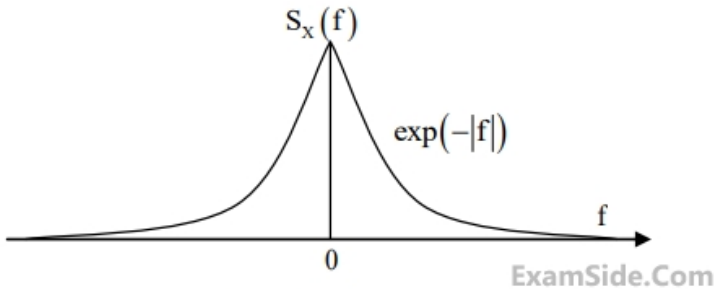


Fig. (a)

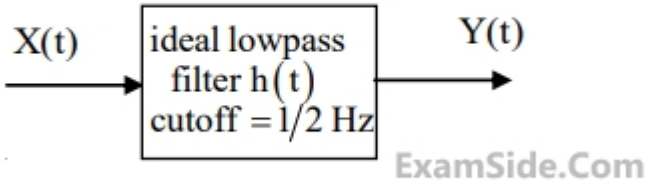


Fig. (b)

Let E be the expectation operator and consider the following statements :

(I) $E(X(t)) = E(Y(t))$

(II) $E(X^2(t)) = E(Y^2(t))$

(III) $E(Y^2(t)) = 2$

Select the correct option:

(A) only I is true

(B) only II and III are true

(C) only I and II are true

(D) only I and III are true

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Q24. GATE ECE 2016 Set 2

Open solution

An information source generates a binary sequence $\{\alpha_n\}$. α_n can take one of the two possible values −1 and +1 with equal probability and are statistically independent and identically distributed. This sequence is pre-coded to obtain another sequence $\{\beta_n\}$, as $\beta_n = \alpha_n + k\alpha_{n-3}$. The sequence $\{\beta_n\}$ is used to modulate a pulse $g(t)$ to generate the baseband signal $x(t) = \sum_{n=-\infty}^{\infty} \beta_n g(t - nT)$, where $g(t) = \begin{cases} 1, & 0 \le t \le T \\ 0 & otherwise \end{cases}$

If there is a null at $f = \frac{1}{3T}$ in the power spectral density of $X(t)$, then k is _____.

Q25. GATE ECE 2016 Set 2

Open solution

Consider random process $X(t) = 3V(t) - 8$, where $V(t)$ is a zero mean stationary random process with autocorrelation $R_v(\tau) = 4e^{-5|\tau|}$. The power of $X(t)$ is _____.

Q26. GATE ECE 2016 Set 1

Open solution

An antenna pointing in a certain direction has a noise temperature of 50K. The ambient temperature is 290K. The antenna is connected to a pre-amplifier that has a noise figure of 2dB and an available gain of 40 dB over an effective bandwidth of 12 MHz. The effective input noise temperature T_e for the amplifier and the noise power P_{ao} at the output of the preamplifier, respectively, are

(A) $T_e = 169.36K$ and $P_{ao} = 3.73 \times 10^{-10}\ W$

(B) $T_e = 170.8K$ and $P_{ao} = 4.56 \times 10^{-10}\ W$

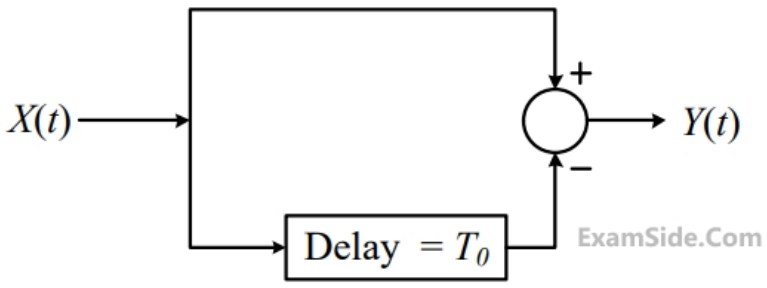
(C) $T_e = 182.5K$ and $P_{ao} = 3.85 \times 10^{-10}\ W$

(D) $T_e = 160.62K$ and $P_{ao} = 4.6 \times 10^{-10}\ W$

Q27. GATE ECE 2016 Set 3

Open solution

A wide sense stationary random process $X(t)$ passes through the LTI system shown in the figure. If the autocorrelation function of $X(t)$ is $R_x(\tau)$, then the autocorrelation function $R_x(\tau)$, of the output $Y(t)$ is equal to



(A) $2R_X(\tau) + R_X(\tau - T_0) + R_X(\tau + T_0)$

(B) $2R_X(\tau) - R_X(\tau - T_0) - R_X(\tau + T_0)$

(C) $2R_X(\tau) + 2R_X(\tau - 2T_0)$

(D) $2R_X(\tau) - 2R_X(\tau - 2T_0)$

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Q28. GATE ECE 2015 Set 3

Open solution

A random binary wave $y(t)$ is given by

$$y(t) = \sum_{n=-\infty}^{\infty} X_n p(t - nT - \phi)$$

where $p(t) = u(t) - u(t-T)$, $u(t)$ is the unit step function and ϕ is an independent random variable with uniform distribution in $[0, T]$. The sequence $\{X_n\}$ consists of independent and identically distributed binary valued random variables with $P\{X_n = +1\} = P\{X_n = -1\} = 0.5$ for each n.

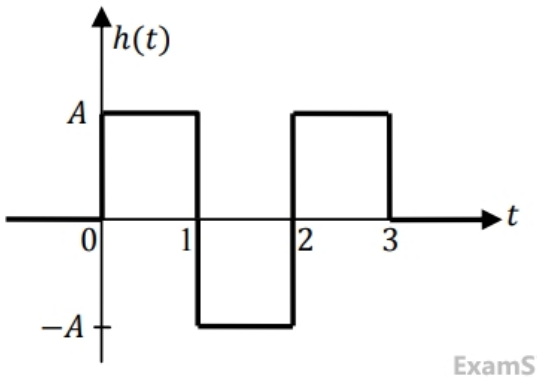
The value of the autocorrelation $R_{yy}\left(\frac{3T}{4}\right) \underline{\underline{\Delta}} E\left[y(t) y\left(t - \frac{3T}{4}\right)\right]$

equals ---------- .

Q29. GATE ECE 2015 Set 2

Open solution

A zero mean white Gaussian noise having power spectral density $\frac{N_0}{2}$ is passed through an LTI filter whose impulse response $h(t)$ is shown in the figure. The variance of the filtered noise at $t = 4$ is



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(A) $\frac{3}{2}A^2N_0$ (B) $\frac{3}{4}A^2N_0$ (C) A^2N_0 (D) $\frac{1}{2}A^2N_0$

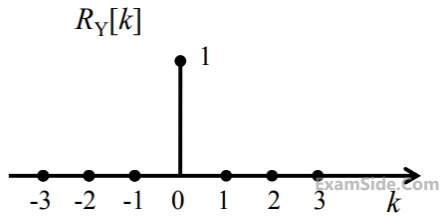
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Q30. GATE ECE 2015 Set 2

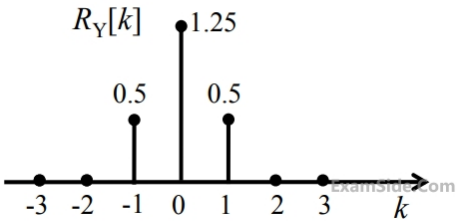
$\{X_n\}_{n=-\infty}^{n=\infty}$ is an independent and identically distributed (i.i.d) random process with X_n equally likely to be $+1$ or -1. $\{Y_n\}_{n=-\infty}^{n=\infty}$ is another random process obtained as $Y_n = X_n + 0.5X_{n-1}$.
The autocorrelation function of $\{Y_n\}_{n=-\infty}^{n=\infty}$, denoted by $r_y[K]$, is

Open solution

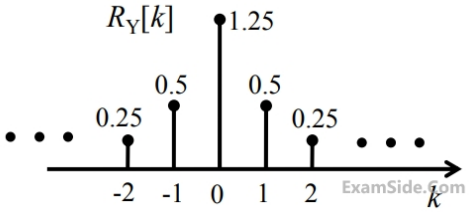
(A)



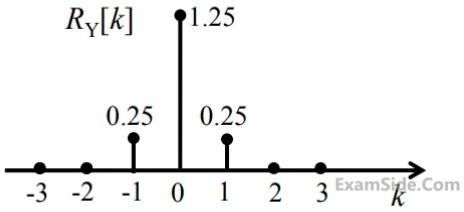
(B)



(C)



(D)



Q31. GATE ECE 2015 Set 2

Let $X \in \{0, 1\}$ and $Y \in \{0, 1\}$ be two independent binary random variables.
If $P(X = 0) = p$ and $P(Y = 0) = q$, then $P(X + Y \geq 1)$ is equal to

Open solution

(A) $pq + (1-p)(1-q)$

(B) pq

(C) $p(1-q)$

(D) $1 - pq$

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Q32. GATE ECE 2014 Set 1

Consider a random process $X(t) = \sqrt{2}\sin(2\pi t + \varphi)$, where the random phase φ is uniformly distributed in the interval $[0,\ 2\pi]$. The auto - correlation $E[X(t_1)X(t_2)]$ is

Open solution

(A) $\cos(2\pi(t_1 + t_2))$

(B) $\sin(2\pi(t_1 - t_2))$

(C) $\sin(2\pi(t_1 + t_2))$

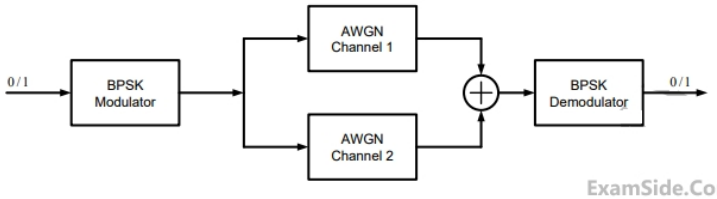
(D) $\cos(2\pi(t_1 - t_2))$

Q33. GATE ECE 2014 Set 1

Open solution

Let $Q(\sqrt{\gamma})$ be the BER of a BPSK system over an AWGN channel with two - sided noise power spectral density $N_0/2$. The parameter γ is a function of bit energy and noise power spectral density.

A system with two independent and identical AWGN channels with noise power spectral density $N_0/2$ is shown in the figure. The BPSK demodulator receives the sum of outputs of both the channels



If the BER of this system is $Q(b\sqrt{\gamma})$, then the value of b is ----------.

Q34. GATE ECE 2014 Set 2

Open solution

The power spectral density of a real stationary random process X(t) is given by

$$S_x(f) = \begin{cases} \frac{1}{W}, |f| \le W \\ 0, |f| > W \end{cases}$$

The value of the expectation

$$E\left[\pi X(t) X\left(t - \frac{1}{4W}\right)\right]$$

is ----------.

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Q35. GATE ECE 2014 Set 3

Open solution

Let $X(t)$ be a wide sense stationary (WSS) random procfess with power spectral density $S_x(f)$. If $Y(t)$ is the process defined as $Y(t) = X(2t-1)$, the power spectral density $S_y(f)$ is .

(A) $S_y(f) = \frac{1}{2} S_x\left(\frac{f}{2}\right) e^{-j\pi f}$

(B) $S_y(f) = \frac{1}{2} S_x\left(\frac{f}{2}\right) e^{-j\pi f/2}$

(C) $S_y(f) = \frac{1}{2} S_x\left(\frac{f}{2}\right)$

(D) $S_y(f) = \frac{1}{2} S_x\left(\frac{f}{2}\right) e^{-j2\pi f}$

Q36. GATE ECE 2014 Set 3

Open solution

A real band-limited random process $X(t)$ has two -sided power spectral density

$$S_x(f) = \begin{cases} 10^{-6}(3000 - |f|) Watts/Hz & for\, |f| \le 3kHz \\ 0 & otherwise \end{cases}$$

Where f is the frequency expressed in Hz. The signal $X(t)$ modulates a carrier cos $16000\,\pi t$ and the resultant signal is passed through an ideal band-pass filter of unity gain with centre frequency of $8kHz$ and band-width of $2kHz$. The output power (in Watts) is _____.

Q37. GATE ECE 2014 Set 3

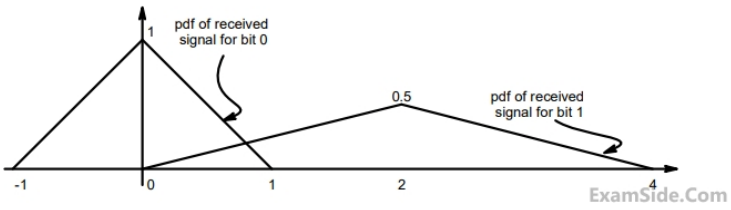
Open solution

Let X_1, X_2, and X_3 be independent and identically distributed random variables with the uniform distribution on $[0, 1]$. The probability $P\{X_1 + X_2 \le X_3\}$ is _____ .

Q38. GATE ECE 2013

Open solution

Bits 1 and 0 are transmitted with equal probability. At the receiver, the pdf of the respective received signals for both bits are as shown below.



If the detection threshold is 1, the BER will be

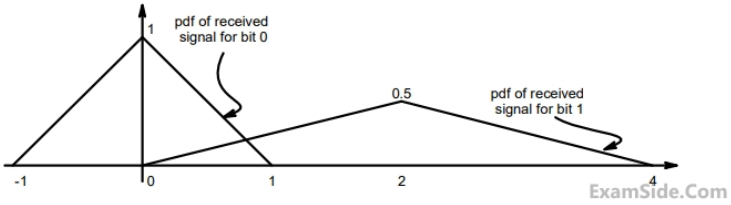
(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{16}$

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Q39. GATE ECE 2013

Bits 1 and 0 are transmitted with equal probability. At the receiver, the pdf of the respective received signals for both bits are as shown below.

Open solution



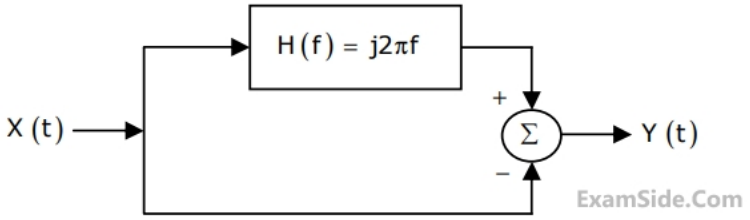
The optimum threshold to achieve minimum bit error rate (BER) is

(A) $\frac{1}{2}$ (B) $\frac{4}{5}$ (C) 1 (D) $\frac{3}{2}$

Q40. GATE ECE 2011

X(t) is a stationary random process with autocorrelation function $R_X(\tau) = \exp(-\pi\tau^2)$. This process is passed through the system shown below. The power spectral density of the output process Y(t) is

Open solution



(A) $\left(4\pi^2 f^2 + 1\right) \exp\left(-\pi f^2\right)$ (B) $\left(4\pi^2 f^2 - 1\right) \exp\left(-\pi f^2\right)$

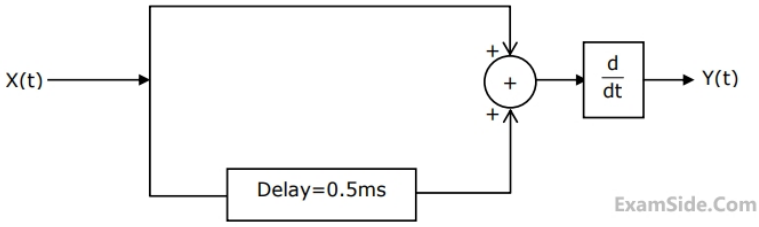
(C) $\left(4\pi^2 f^2 + 1\right) \exp\left(-\pi f\right)$ (D) $\left(4\pi^2 f^2 - 1\right) \exp\left(-\pi f\right)$

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Q41. GATE ECE 2010

X(t) is a stationary process with the power spectral density $S_x(f) > 0$ for all f. The process is passed through a system shown below.

Open solution



Let $S_y(f)$ be the power spectral density of Y(t). Which one of the following statements is correct?

(A) $S_y(f) > 0$ for all f

(B) $S_y(f) > 0$ for $|f| > 1$ kH

(C) $S_y(f) > 0$ for $f = nf_0$, $f_0 = 2$kHz, n any integer

(D) $S_y(f) > 0$ for $f = (2n + 1)f_0$, $f_0 = 1$ kHz, n any integer

Q42. GATE ECE 2008

Noise with double-sided power spectral density of K over all frequencies is passed through a RC low pass filter with 3-dB cut-off frequency of f_c. The noise power at the filter output is

Open solution

(A) K (B) $K f_c$ (C) $K \pi f_c$ (D) ∞

Q43. GATE ECE 2006

A zero-mean white Gaussian noise is passed through an ideal low-pass filter of bandwidth 10 kHz. The output is then uniformly sampled with sampling period $t_s = 0.03$ msec. The samples so obtained would be

Open solution

(A) correlated (B) statistically independent

(C) uncorrelated (D) orthogonal

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Q44. GATE ECE 2006

Open solution

The following question refer to wide sense stationary stochastic process:
It is desired to generate a stochastic process (as voltage process) with power spectral density

$$S(\omega) = \frac{16}{16 + \omega^2}$$

By driving a Linear-Time-Invariant system by zero mean white noise (as voltage process) with power spectral density being constant equal to 1. The system which can perform the desired task could be

(A) first order lowpass R-L filter
(B) first order highpass R-c filter
(C) tuned L-C filter
(D) series R-L-C filter

Q45. GATE ECE 2006

Open solution

The following question refer to wide sense stationary stochastic process:
The parameters of the system obtained in Q. 12 would be

(A) first order R-L low pass filter would have $R = 4\Omega\, L = 1\, H$
(B) first order R-C high pass filter would have $R = 4\Omega\, C = 0.25F$
(C) tuned L-C filter would have $L = 4H\ C = 4F$.
(D) series R-L-C low pass filter would have $R = 1\Omega,\ L = 4H,\ C = 4F$.

Q46. GATE ECE 2005

Open solution

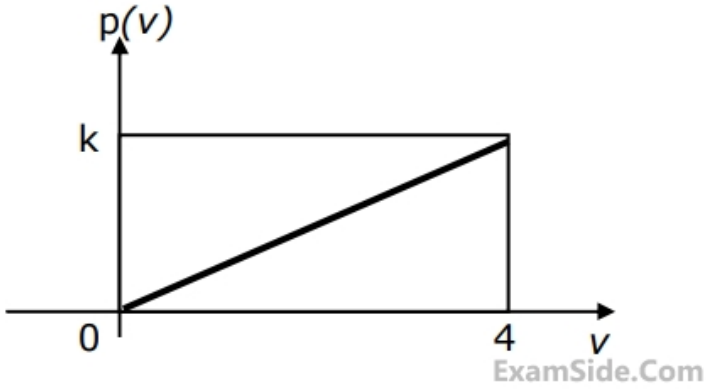
Noise with uniform power spectral density of N_0 W/Hz is passed through a filter H(ω) = 2exp (-jωt_d) followed by an ideal low pass filter of bandwidth B Hz. The output noise power in Watts is

(A) $2N_0B$ (B) $4\ N_0B$ (C) $8N_0B$ (D) $16N_0B$

Q47. GATE ECE 2005

Open solution

An output of a communication channel is a random variable 'V' with the probability density function as shown in the figure. The mean square value of 'V' is



(A) 4 (B) 6 (C) 8 (D) 9

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Q48. GATE ECE 2004

A 1 mW video signal having a bandwidth of 100 MHz is transmitted to a receiver through a cable that has 40 dB loss. If the effective one-sided noise spectral density at the receiver is 10^{-20} Watt/Hz, then the signal-to-noise ratio at the receiver is

(A) 50 dB (B) 30 dB (C) 40 dB (D) 60 dB

Open solution

Q49. GATE ECE 2002

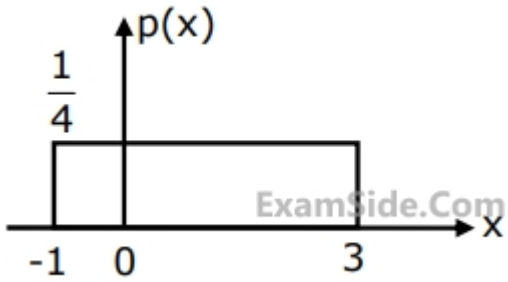
If the variance σ_d^2 of d(n) = x(n) - x(n - 1) is one-tenth the variance σ_x^2 of a stationary zero-mean discrete-time signal x(n), then the normalized autocorrelation function $R_{xx}(k) / \sigma_x^2 \, at \, k = 1$ is

(A) 0.95 (B) 0.90 (C) 0.10 (D) 0.05

Open solution

Q50. GATE ECE 1992

For a random variable 'X' following the probability density function, p (x), shown in figure, the mean and the variance are, respectively.



(A) 1/2 and 2/3 (B) 1 and 4/3 (C) 1 and 2/3 (D) 2 and 4/3

Open solution

Q51. GATE ECE 1991

Two resistors R_1 and R_2 (in ohms) at temperatures $T_1{}^\circ K$ and $T_2{}^\circ K$ respectively, are connected in series. Their equivalent noise temperature is.

Open solution

Q52. GATE ECE 1989

Zero mean Gaussian noise of variance N is applied to a half wave rectifier. The mean squared value of the rectifier output will be:

(A) Zero (B) N/2 (C) $N/\sqrt{2}$ (D) N

Open solution

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Q53. GATE ECE 1989

A part of a communication system consists of an amplifier of effective noise temperature, $Te = \ 21\ ^{\circ}K$, and a gain of 13 dB, followed by a cable with a loss of 3 dB. Assuming the ambient temperature to be $300^{\circ}\ K$, we have for this part of the communication system,

(A) effective noise temperature $= 30^{\circ}\ K$

(B) effective noise temperature $= 36^{\circ}\ K$

(C) noise figure = 0.49 dB

(D) noise figure = 1.61 dB

Open solution

Q54. GATE ECE 1987

White Gaussian noise is passed through a linear narrow band filter. The probability density function of the envelope of the noise at the filter output is:

(A) Uniform (B) Poisson (C) Gaussian (D) Rayleigh

Open solution

Q55. GATE ECE 1987

The variance of a random variable X is σ_x^2 . Then the variance of - kx (where k is a positive constant) is

(A) σ_x^2 (B) - k σ_x^2 (C) k σ_x^2 (D) $k^2\ \sigma_x^2$

Open solution

Marks 4 1 questions

Q56. GATE ECE 1988

White Gaussian noise with zero mean and double - sided power spectral density $\eta/2$ is the input $x(t)$ to a linear system with impulse response $h(t) = exp\left(-t/RC\right) \quad u\left(t\right)$. The output is $y(t)$. Evaluate $E\left[x\left(t+\tau\right)y\left(t\right)\right]$ for $\tau > 0$. (Note: $u(t)$ is the unit step function)

Open solution

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Answer Key

Marks 1

1. **A**	2. **A, B**	3. **A**	4. **2**	5. **6**
6. **2**	7. **A**	8. **B**	9. **B**	10. **A**
11. **C**	12. **C**	13. **D**	14. **C**	15. **A**
16. **D**				

Marks 2

17. **A**	18. **B**	19. **45 to 45.30**	20. **B**	21. **0.24 to 0.26**
22. **1.167**	23. **A**	24. **-1**	25. **100**	26. **A**
27. **B**	28. **0.25**	29. **A**	30. **B**	31. **D**
32. **D**	33. **1.414**	34. **4**	35. **C**	36. **2.5**
37. **0.17**	38. **D**	39. **B**	40. **A**	41. **D**
42. **C**	43. **A**	44. **A**	45. **A**	46. **B**
47. **C**	48. **A**	49. **A**	50. **B**	51. $Te = \frac{R_1T_1 + R_2T_1}{R}\ ^\circ K$
52. **B**	53. **B, C**	54. **D**	55. **D**	

Marks 4

56. **Solve it.**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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